

1. Administrative & Study Identification

Full Study Title: A Randomized, Double-blind, Parallel Group, Multi-center, Phase III Study to Assess the Efficacy of Budesonide, Glycopyrronium, and Formoterol Fumarate Metered Dose Inhaler Relative to Glycopyrronium and Formoterol Fumarate MDI on Cardiopulmonary Outcomes in Chronic Obstructive Pulmonary Disease
Short Title/Acronym: THARROS **Protocol Number:** D5989C00001 **NCT Number:** NCT06283966 **Sponsor Name:** AstraZeneca **Investigational Product:** Budesonide, Glycopyrronium, and Formoterol Fumarate Metered Dose Inhaler (BGF MDI) **Phase of Development:** Phase III **Medical Monitor:** AstraZeneca Clinical Operations Lead

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To evaluate the effect of combination triple therapy with ICS/LAMA/LABA (BGF MDI 320/14.4/9.6 µg) on combined cardiopulmonary outcomes relative to dual LAMA/LABA therapy (GFF MDI 14.4/9.6 µg) in a population with COPD and elevated cardiopulmonary risk. **Secondary Objectives:** To evaluate the effect of BGF MDI compared with GFF MDI on the time to severe COPD exacerbations, time to Myocardial Infarction (MI) hospitalization or cardiac death, and time to Heart Failure (HF) acute healthcare visit/hospitalization or cardiac death.

2.2 Background & Rationale COPD and cardiovascular disease (CVD) are leading causes of death with overlapping and syndemic pathophysiological interactions. Lung damage from COPD leads to deterioration in lung function, flare-ups, and air trapping, which puts more strain on the heart. While inhaled triple therapies reduce COPD exacerbations and improve lung function versus dual therapy, their prospective effect on combined severe cardiac and pulmonary (cardiopulmonary) events in people not currently on maintenance ICS therapies has not been thoroughly tested. THARROS addresses this gap by evaluating if BGF triple therapy can prevent combined COPD flare-ups and serious cardiac events in a high-risk population.

3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional **Control Type:** Active Comparator (GFF MDI 14.4/9.6 µg) **Blinding:** Double-blind **Randomization:** Randomized, Parallel Group, Event-driven trial

3.2 Planned Enrollment & Duration Target \$\$\$: Approximately 5,000 evaluable participants **Number of Sites:** Multicenter,

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Age ≥ 40 to ≤ 80 years old. 2. Confirmed diagnosis of Chronic Obstructive Pulmonary Disease (COPD) with a history of current or former smoking. 3. Elevated cardiopulmonary risk defined by a combination of cardiovascular risk factors (e.g., Hypertension, Diabetes Mellitus, Chronic Kidney Disease, Dyslipidemia, Obesity).

4.2 Exclusion Criteria 1. Active diagnosis of asthma within the past 5 years or asthma-COPD overlap. 2. Previous use of ICS-containing maintenance medication in the 12 months prior to the study. 3. Unstable or life-threatening cardiac arrhythmia requiring intervention in the last 8 weeks, or an MI/unstable angina in the last 8 weeks. 4. End-stage renal disease requiring renal replacement therapy or history/active listing of heart or lung transplant.

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: - Experimental Arm: BGF MDI 320/14.4/9.6 μg , administered twice daily (BID). - Active Comparator Arm: GFF MDI 14.4/9.6 μg , administered twice daily (BID). **Route of Administration:** Inhaled (Metered Dose Inhaler) **Duration of Treatment:** Event-driven duration (continuous treatment until trial concludes or target number of cardiopulmonary events is reached).

5.2 Concomitant & Prohibited Therapy Permitted: Standard-of-care medications for COPD (non-ICS) and established cardiovascular comorbidities. **Prohibited:** Concurrent use of any other inhaled corticosteroids (ICS) or off-protocol LAMA/LABA maintenance therapies during the trial.

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures
Screening	Day -14 to 0	Informed Consent, Medical & Exacerbation History, Cardiopulmonary Risk Assessment, PFTs
Baseline	Day 0	Randomization, Inhaler Technique Training, First Dose
Interim	Event-Driven	Minimal on-site visits supplemented by virtual/remote visits. Direct-to-home study medication shipping (per local regulations), Safety & Event Tracking
End of Study	Target Event Met	Final Vital Signs, Exit Interview, IP Accountability, Final Cardiopulmonary Assessment

7. Outcome Measures (Endpoints)

7.1 Primary Endpoint Definition: Time to first severe cardiopulmonary event (a composite of cardiac-related deaths, severe COPD-related hospitalizations/deaths, and specific severe cardiac events). **Measurement Method:** Clinical Endpoint Committee (CEC) independent adjudication.

7.2 Secondary & Exploratory Endpoints - Time to first severe COPD exacerbation. - Time to MI hospitalization or cardiac death. - Time to Heart Failure (HF) acute healthcare visit/hospitalization or cardiac death.

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Standard Phase III collection via remote visits, on-site assessments, and virtual check-ins.

Serious Adverse Events (SAEs): Specific focus on capturing severe cardiopulmonary deteriorations (MI, HF, pneumonia, severe exacerbations). **Data Monitoring Committee (DMC):** Independent Clinical Endpoint Committee (CEC) tasked with adjudicating cardiac and pulmonary events and mortality causes.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Intent-to-Treat (ITT) for primary efficacy; Safety Set for all treatment-emergent AEs. **Sample Size**

Justification: Target $N = 5,000$ participants properly powered based on expected incidence rates of combined cardiopulmonary events in the high-risk COPD phenotype. **Handling of Missing Data:** Time-to-event analysis (e.g., Cox proportional hazards models) utilizing censoring methods for incomplete participant follow-ups.

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki.

Monitoring Plan: A hybrid monitoring model combining limited on-site visits with virtual/remote monitoring to lower patient burden and increase retention. **Audit Trail:** Data entered into a validated electronic data capture (EDC) system with full audit trails and query generation.

11. Study Identifiers & Governance

Primary ID: D5989C00001 **Registry Number:** NCT06283966 **Sponsor/Lead Organization:** AstraZeneca **Study Title:** THARROS Official

Regulatory Title: A Randomized, Double-blind, Parallel Group, Multi-center, Phase III Study to Assess the Efficacy of Budesonide, Glycopyrronium, and Formoterol Fumarate Metered Dose Inhaler Relative to Glycopyrronium and Formoterol Fumarate MDI on Cardiopulmonary Outcomes in Chronic Obstructive Pulmonary Disease

12. Clinical Framework (Cardiopulmonary Specific)

Primary Condition: Chronic Obstructive Pulmonary Disease (COPD) with elevated CV risk **Diagnosis Threshold:** Clinically diagnosed COPD, aged 40-80, current/former smoker, with specific combinations of cardiovascular comorbidities **Medical Methodology:** Optimization of inhaled maintenance therapy via ICS/LAMA/LABA triple therapy **Optimization Target:** Reduction in the incidence of combined severe cardiac and pulmonary (cardiopulmonary) events

13. Study Procedures & Methodology

Management Pathway: Hybrid trial design incorporating direct-to-patient medication shipments and virtual telehealth visits
Education Protocol (HCP): Site Initiation Visits focusing on cardiopulmonary endpoint identification and CEC reporting
Education Protocol (Patient): Proper MDI inhaler technique and symptom recognition for severe exacerbations/cardiac events
Quality Audit Mechanism: Independent Clinical Endpoint Committee (CEC) adjudication of all severe/fatal events to distinguish cardiac vs. pulmonary etiology

14. Observation & Follow-Up Schedule

Total Duration: Event-driven (estimated ~4 years or 48 months follow-up) **Onsite Visit Cadence:** Minimal essential onsite visits required by protocol to ensure patient retention **Remote Monitoring Frequency:** Hybrid virtual visits throughout the event-driven period **Checklist Review Interval:** Routine symptom, hospitalization, and exacerbation checklist review conducted during remote check-ins

15. Sign-off & Approval

Role	Name	Signature	Date	:--	:--	:--	:--
Principal Investigator	[Name]	_____	[DD-MMM-YYYY]				
Clinical Operations Lead	[Name]	_____	[DD-MMM-YYYY]		Lead		
Statistician	[Name]	_____	[DD-MMM-YYYY]				